CAMPUS EVENT FINANCIAL PLANNING

Python Program – Complete Assignment Documentation

1. Problem Statement

Develop a Python program to calculate the budget and financial status of a campus event. The program accepts fixed expenses (venue, equipment and transport), per-person expenses (catering and certificate), participant count, contribution per participant and sponsorship. It calculates total costs, available funds, funding shortfall and the minimum registrations needed to break even.

2. Assumptions

• All monetary inputs are non-negative and use the same currency.

• The number of participants is a non-negative whole number.

• Catering and certificate costs apply to each participant.

• Venue, equipment and transport are fixed costs.

• Each participant pays the same contribution; sponsorship is available funding.

• The break-even calculation assumes fixed costs and sponsorship do not change with registrations.

3. Algorithm

1. Start the program.

2. Input venue, equipment, catering, transport and certificate costs.

3. Input participants, contribution per participant and sponsorship.

4. Calculate fixed cost = venue + equipment + transport.

5. Calculate variable cost = participants × (catering + certificate).

6. Calculate total event cost = fixed cost + variable cost.

7. Calculate participant funds = participants × contribution.

8. Calculate total funds = participant funds + sponsorship.

9. Calculate shortfall = max(0, total event cost – total funds).

10. Calculate balance per person = contribution – catering – certificate.

11. If balance per person is positive, calculate minimum registrations by rounding up (fixed cost – sponsorship) / balance per person, with a minimum of zero.

12. Otherwise, display that break-even is not possible.

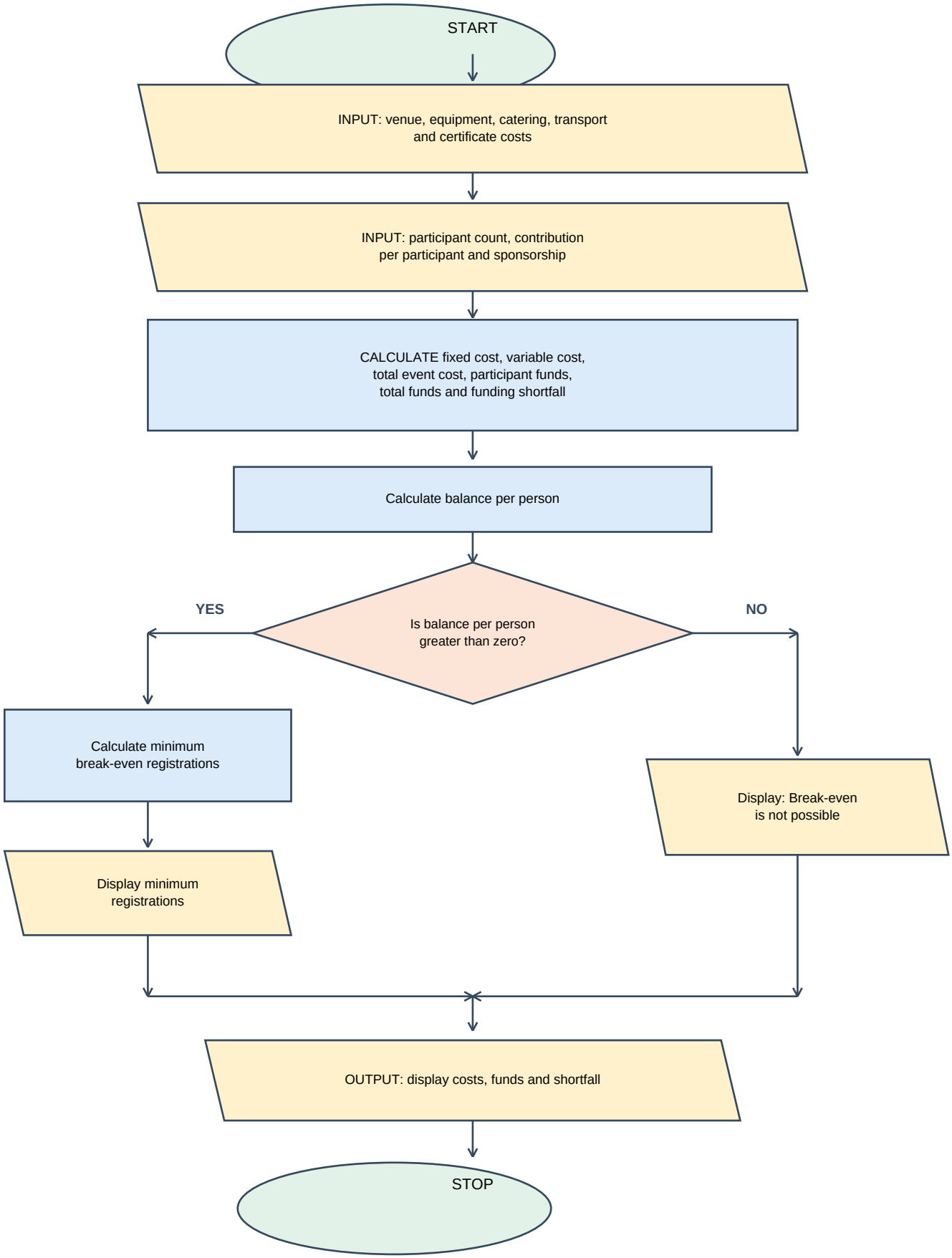
13. Display all costs, funds, shortfall and break-even result. Stop.

4. Pseudocode

```
START
    INPUT venue, equipment, catering, transport, certificate
    INPUT participants, contribution, sponsorship

    fixed_cost = venue + equipment + transport
    variable_cost = participants * (catering + certificate)
    total_cost = fixed_cost + variable_cost

    participant_funds = participants * contribution
    total_funds = participant_funds + sponsorship
    shortfall = MAX(0, total_cost - total_funds)

    balance_per_person = contribution - catering - certificate

    IF balance_per_person > 0 THEN
        minimum = MAX(0, CEILING((fixed_cost - sponsorship)
                                 / balance_per_person))
        PRINT minimum registrations to break even
    ELSE
        PRINT "Break-even is not possible"
    END IF

    PRINT fixed_cost, variable_cost, total_cost
    PRINT participant_funds, sponsorship, shortfall
STOP
```

5. Flowchart with Standard Symbols

Oval = Start/Stop | Parallelogram = Input/Output | Rectangle = Process | Diamond = Decision | Arrow = Flow direction



6. Sample Input and Output

Sample values (all amounts in rupees): venue = 10000, equipment = 5000, catering per person = 200, transport = 3000, certificate per person = 50, participants = 100, contribution per participant = 400, sponsorship = 10000.

```
Minimum registrations to break even: 54
Fixed Cost: 18000.0
Variable Cost: 25000.0
Total Event Cost: 43000.0
Participant Contributions: 40000.0
Sponsorship: 10000.0
Sponsorship/Funding Shortfall: 0
```

Calculation: fixed cost = 18,000; variable cost = 25,000; total cost = 43,000; total funds = 50,000; shortfall = 0. The balance per person is 400 – 200 – 50 = 150. After sponsorship, fixed cost to cover is 18,000 – 10,000 = 8,000. Ceiling(8,000 / 150) = 54 registrations.

7. Python Program

```
# Campus Event Financial Planning

venue = float(input("Enter venue cost: "))
equipment = float(input("Enter equipment cost: "))
catering = float(input("Enter catering cost per person: "))
transport = float(input("Enter transport cost: "))
certificate = float(input("Enter certificate cost per person: "))

participants = int(input("Enter number of participants: "))
contribution = float(input("Enter contribution per participant: "))
sponsorship = float(input("Enter sponsorship amount: "))

# Calculate total cost
fixed_cost = venue + equipment + transport
variable_cost = participants * (catering + certificate)
total_cost = fixed_cost + variable_cost

# Calculate total funds
participant_funds = participants * contribution
total_funds = participant_funds + sponsorship

# Calculate shortfall
shortfall = max(0, total_cost - total_funds)

# Minimum registrations to break even
balance_per_person = contribution - catering - certificate

if balance_per_person > 0:
    minimum = max(0, int((fixed_cost - sponsorship) / balance_per_person))
    if minimum * balance_per_person < fixed_cost - sponsorship:
        minimum += 1
    print("Minimum registrations to break even:", minimum)
else:
    print("Break-even is not possible with the given contribution.")

print("Fixed Cost:", fixed_cost)
print("Variable Cost:", variable_cost)
print("Total Event Cost:", total_cost)
print("Participant Contributions:", participant_funds)
print("Sponsorship:", sponsorship)
print("Sponsorship/Funding Shortfall:", shortfall)
```

8. Conclusion

The program calculates the total campus event expenses and compares them with participant contributions and sponsorship. It finds the funding shortfall and the minimum registrations required to cover fixed costs after sponsorship, when each participant contributes more than their variable cost. This helps event organizers plan the budget and check whether the event can break even.